MIRTHIKHA N S

B. Tech AI & Data Science • Machine Learning • Computer Vision

Coimbatore, Tamil Nadu • mirthi1303@gmail.com • +91-8778600868 • linkedin.com/in/mirthikha-n-s • github.com/Mirthikha

PROFESSIONAL SUMMARY

A 3rd year AI & Data Science engineering student with hands-on experience in deep learning, computer vision, and predictive modeling. Experienced in deploying and building CNN architectures, executing image and data preprocessing pipelines and developing interactive end-to-end data applications using Streamlit. Proficient in Python, TensorFlow, Scikit-learn, and OpenCV. Experienced in writing clean, structured code in Python, Java, and C, alongside a strong core technical knowledge of SQL database concepts.

EDUCATION

B. Tech in Artificial Intelligence & Data Science 2024 – 2028

V.S.B College of Engineering Technical Campus, Coimbatore | CGPA: **9.04/ 10.0**

PROFESSIONAL EXPERIENCE

Machine Learning Intern | Prodigy InfoTech

Remote | June 1, 2025 to June 30, 2025

- **Earned an official Letter of Recommendation (LOR)** for rapid learning ability and meeting all deadlines efficiently.
- Built a **hand gesture recognition system** using TensorFlow, **SVM Image Classifier, House Price Prediction** using Linear Regression.

Machine Learning Intern | Profenaa Technologies

Remote | June 2025 to July 2025

- **Developed a Salary Prediction Script:** Implemented a scikit-learn linear regression and logistic regression model to predict continuous expected_salary targets based on features.
- **Conducted Data Feature Visualization:** Used matplotlib to generate scatter plots, trend lines, and box plots, identifying feature distributions that improved regression fit.

Machine Learning Intern | Sysslan IT Solutions

Remote | December 2025 to January 2026

Train Journey Duration Predictor | *Python • Scikit-learn • Pandas • Streamlit*

Code repository: github.com/Mirthikha/train-journey-duration-prediction

- Built an end-to-end ML pipeline using Linear Regression to predict train journey durations with an interactive Streamlit web application enabling users to input journey parameters and receive real-time duration predictions.
- Trained 3 Linear Regression models with different features and finalized a model with higher accuracy.
- **Evaluated** pipeline performance to achieve an **R square score of 0.698**, successfully mapping approximately **70%** of variance within the transit scheduling data.

Artificial Intelligence Intern – Codsoft

Remote | June 1, 2026 to June 30, 2026

University Admission Assistant Chatbot | Python • Django

- Developed a Rule-Based University Chatbot: Built a chatbot using Python and Django that uses conditional logic to instantly answer common questions about university admissions.
- Connected Frontend to Backend: Used HTML, CSS, and JavaScript to create a simple chat interface that sends user messages to Django and displays replies in real time.

Image Captioning AI | Python • CNN • Transformer • ResNet

- Built an image captioning system that extracts visual features using a pre-trained ResNet model and generates natural language descriptions with an LSTM decoder.

Tic-Tac-Toe Game | Python • Flask • MiniMax Algorithm

- Built an Unbeatable Tic-Tac-Toe AI: Implemented the Minimax algorithm with backtracking in Python and Flask for frontend.

PROJECTS

CivicMind – AI-Powered Multi-Agent Civic Intelligence Platform | *Python • Gemini API • Firebase • Multi - Agent • FastAPI*

Website Link: https://civicmind-ai-platform.onrender.com

- Built a multi-agent pipeline using FastAPI and Google Gemini that routes city complaints to the officials, estimates resolution budgets, generates explainable AI justifications for each decision, and assigns SLA resolution targets.
- Developed a self-correction layer that audits raw LLM outputs and auto-heals schema errors before writing to Firebase, keeping the UI completely crash-free.
- Built a classification flow that detects life-threatening emergency reports instantly and triggers high-priority alerts on official dispatch boards.
- Designed an executive dashboard with charts to visualize and cluster related complaints sharing an underlying root cause (e.g., water pipe leaks causing road damage).

Intelligent Candidate Ranking System | *Python • Sentence Transformer • Vector Embeddings • Cosine Similarity • React*

Website Link: https://intelligent-candidate-discovery-ranking-system-3jlsjnab3zbntuu.streamlit.app

- Built an end-to-end resume scoring engine that transforms candidate profiles and job descriptions into semantic text embeddings, ranking top applicants using cosine similarity.
- Filtered out invalid resumes, spam applications, and trap "honeypots" by spotting red-flag signs (Tier-5 unverified institutions, suspicious metadata, and impossible skill claims).
- Engineered the entire system to run 100% offline (zero internet/external API access) within a strict 5-minute execution window, leveraging multi-process CPU chunking and local model caching to process high-volume candidate batches under extreme compute constraints.
- Used MinHash and Locality-Sensitive Hashing (LSH) to spot and eliminate mass-copied resumes and template clones in milliseconds based on text overlap Applied custom behavioral multipliers to prioritize verified, high-intent job seekers.

Driver Drowsiness Detection System | *Python • TensorFlow • CNN • OpenCV*

Code repository: github.com/Mirthikha/Driver-drowsiness-detection

- Achieved 99.95% validation accuracy on static image datasets; identified and diagnosed a real-world distribution shift when deploying to live video (58.52% accuracy), motivating a temporal post-processing fix.
- Implemented temporal post-processing pipeline to reduce false-positive alerts by smoothing frame-level predictions, improving system reliability for real-world deployment.

SKILLS

Languages: Python, Java, SQL, C, HTML, CSS, Bootstrap

ML / DL Frameworks: TensorFlow, Keras, Scikit-learn, OpenCV

Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn

Deployment & Tools: Streamlit, GitHub, VS Code, Google Colab, Jupyter Notebook, PyCharm

Core Concepts: CNNs, Linear Regression, Logistic Regression, SVM, K-Means Clustering, Feature Engineering, Data Preprocessing, Model Evaluation (RMSE, R^2), Confusion Matrix

Soft Skills: Technical Documentation

CERTIFICATIONS & ACHIEVEMENTS

Shortlisted among the **Top 100** finalists across the **APAC** region in the **Google Cloud Gen AI Hackathon (Cohort 2)** for developing **CivicMind** – AI-Powered Multi-Agent Civic Intelligence Platform.

Infosys Springboard - Natural Language Processing (NLP), Deep Learning, Generative AI, Computer Vision 101, Robotic Process Automation, Python Data Structures & Algorithms

Academic Achievement - Attained a Centum in Computer Science during Class XI Board Examinations.

National Conference - Presented a research paper "Multi-Modal Journey Optimization" at a national-level academic conference at V.S.B College of Engineering Technical campus

Hackathon - KPR Inter-College Hackathon, Google Cloud Gen AI Hackathon, India Runs Hackathon